How much can structure tell us?

Two mysteries for Groups, Rings and Fields

MTH2010

If I tell you that a group has six elements, have I told you which group it is?

$\mathbb{Z}_6$	$\neq$	D_6
$0, 1, 2, 3, 4, 5$		six symmetries of a triangle
addition modulo 6		composition of movements

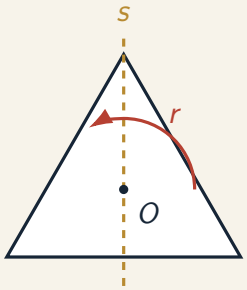
Here D_6 means the triangle's symmetry group, which has six elements.

Two groups with six elements

$\mathbb{Z}_6$ under addition

$+$	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$
$[0]_6$	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$
$[1]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$	$[0]_6$
$[2]_6$	$[2]_6$	$[3]_6$	$[4]_6$	$[5]_6$	$[0]_6$	$[1]_6$
$[3]_6$	$[3]_6$	$[4]_6$	$[5]_6$	$[0]_6$	$[1]_6$	$[2]_6$
$[4]_6$	$[4]_6$	$[5]_6$	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$
$[5]_6$	$[5]_6$	$[0]_6$	$[1]_6$	$[2]_6$	$[3]_6$	$[4]_6$

D_6: symmetries of a triangle



r: anticlockwise rotation through $2\pi/3$
s: reflection in the dashed axis

An essential difference

In $\mathbb{Z}_6$

$$a + b = b + a$$

The order never matters.

In D_6

$$rs \neq sr$$

Reflect, then rotate gives a different movement from rotate, then reflect.

Essentially different

No renaming of the elements can turn one multiplication table into the other.

What changes if a group has exactly fifteen elements?

$$|G| = 15$$

We don't know the elements. We don't know the operation. We don't have a table.

The surprising answer

Up to renaming its elements, there is only one possible group.

A glimpse of the order-15 argument

$$15 = 3 \cdot 5$$

The factors 3 and 5 force the group to contain tightly constrained pieces of those sizes.

Those pieces must fit together in only one way. The group contains an element

$$g, \quad g^2, \quad g^3, \quad \ldots, \quad g^{15} = e$$

whose powers run through all fifteen elements.

Later, the Sylow theorems will turn the arithmetic of 15 into information about G.

Increasingly powerful ways to understand finite groups

During the first part of this module, we will develop increasingly powerful ways of understanding finite groups.

Eventually, the Sylow theorems will let us extract an extraordinary amount of information about a group simply from the number of elements it contains.

They will give us the tools to explain why six elements allow genuinely different groups, while fifteen elements allow only one, up to isomorphism.

How much information is hidden in $|G|$?

The Fundamental Theorem of Arithmetic

Every integer greater than 1 factors into primes,

and the factorisation is unique apart from order.

$$60 = 2^2 \cdot 3 \cdot 5$$

Why does this work? Which properties of the integers make it work?

A slightly larger number system

$$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$$

We add and multiply in the usual way, using

$$(\sqrt{-5})^2 = -5.$$

For example,

$$(1 + \sqrt{-5})(1 - \sqrt{-5}) = 1 - (-5) = 6.$$

The rules look familiar. The arithmetic behaves differently.

The second mystery

$$6 = 2 \cdot 3$$

$$= (1 + \sqrt{-5})(1 - \sqrt{-5})$$

Each of

$$2, \qquad 3, \qquad 1 + \sqrt{-5}, \qquad 1 - \sqrt{-5}$$

is irreducible: it cannot be broken into genuinely smaller factors in this number system.

Unique factorisation has failed

The two factorisations remain different after reordering and multiplying factors by ± 1.

How can we know the factors are irreducible?

Attach a non-negative integer to each element:

$$N(a + b\sqrt{-5}) = a^2 + 5b^2.$$

The key property is

$$N(xy) = N(x)N(y).$$

A non-trivial factorisation of our four elements would force an element of norm 2 or 3. But neither equation

$$a^2 + 5b^2 = 2, \qquad a^2 + 5b^2 = 3$$

has an integer solution.

A simple numerical invariant exposes the hidden algebraic structure.

Prime and irreducible come apart

In the integers, we often use these words as though they meant the same thing. Here they do not.

The element 2 is irreducible, yet

$$2 \mid (1+\sqrt{-5})(1-\sqrt{-5}) = 6,$$

while

$$2 \nmid (1+\sqrt{-5}) \qquad \text{and} \qquad 2 \nmid (1-\sqrt{-5}).$$

The question behind the ring theory

What conditions make irreducible elements behave like prime elements, and hence restore unique factorisation?

Where the module is going

Finite groups

$$|G| = 15$$

The Sylow theorems reveal how the size of a group constrains its structure.

Rings and factorisation

$$\text{ED} \Rightarrow \text{PID} \Rightarrow \text{UFD}$$

In a UFD, prime and irreducible elements coincide.

By the end, both mysteries will have answers.